

IP Addressing and Subnetting

Workbook
Version 1.5

11111110

10010101

00011011

10000110

11010011

Student Name:

Address Class Identification

Address	Class
10.250.1.1	<u>A</u>
150.10.15.0	<u>B</u>
192.14.2.0	<u> </u>
148.17.9.1	<u> </u>
193.42.1.1	<u> </u>
126.8.156.0	<u> </u>
220.200.23.1	<u> </u>
230.230.45.58	<u> </u>
177.100.18.4	<u> </u>
119.18.45.0	<u> </u>
249.240.80.78	<u> </u>
199.155.77.56	<u> </u>
117.89.56.45	<u> </u>
215.45.45.0	<u> </u>
199.200.15.0	<u> </u>
95.0.21.90	<u> </u>
33.0.0.0	<u> </u>
158.98.80.0	<u> </u>
219.21.56.0	<u> </u>

Network & Host Identification

Circle the network portion
of these addresses:

177.100.18.4

119.18.45.0

209.240.80.78

199.155.77.56

117.89.56.45

215.45.45.0

192.200.15.0

95.0.21.90

33.0.0.0

158.98.80.0

217.21.56.0

10.250.1.1

150.10.15.0

192.14.2.0

148.17.9.1

193.42.1.1

126.8.156.0

220.200.23.1

Circle the host portion of
these addresses:

10.15.123.50

171.2.199.31

198.125.87.177

223.250.200.222

17.45.222.45

126.201.54.231

191.41.35.112

155.25.169.227

192.15.155.2

123.102.45.254

148.17.9.155

100.25.1.1

195.0.21.98

25.250.135.46

171.102.77.77

55.250.5.5

218.155.230.14

10.250.1.1

Custom Subnet Masks

Problem 1

Number of needed subnets **14**
 Number of needed usable hosts **14**
 Network Address **192.10.10.0**

Address class C

Default subnet mask 255 . 255 . 255 . 0

Custom subnet mask 255 . 255 . 255 . 240

Total number of subnets 16

Total number of host addresses 16

Number of usable addresses 14

Number of bits borrowed 4

Show your work for Problem 1 in the space below.

	256	128	64	32	16	8	4	2	1	Number of Hosts
Number of Subnets	-	2	4	8	16	32	64	128	256	
	128	64	32	16	8	4	2	1	-	Binary values
192 . 10 . 10 . 0	0	0	0	0	0	0	0	0	0	

Add the binary value numbers to the left of the line to create the custom subnet mask.

128
64
32
+16
240

16	Observe the total number of hosts.
-2	
14	Subtract 2 for the number of usable hosts.

Custom Subnet Masks

Problem 4

Number of needed subnets **6**

Number of needed usable hosts **30**

Network Address **210.100.56.0**

Address class _____

Default subnet mask _____

Custom subnet mask _____

Total number of subnets _____

Total number of host addresses _____

Number of usable addresses _____

Number of bits borrowed _____

Show your work for Problem 4 in the space below.

		Number of Subnets					Number of Hosts					
		256	128	64	32	16	8	4	2	-		
	-	2	4	8	16	32	64	128	256			
		128	64	32	16	8	4	2	1	-	Binary values	
210	.	100	.	56	.	0	0	0	0	0		